Assignment 5: JavaScript Basics

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GitHub: https://github.com/mmionya/assignment-5-web

Task 1 - Option A: Higher or Lower

The game generates integers from 1 to 20, including both ends. A correct guess increases the streak; an equal value also counts as correct. A wrong guess removes one of three lives. At zero lives, guessing stops. Restart resets the round and keeps the best streak.

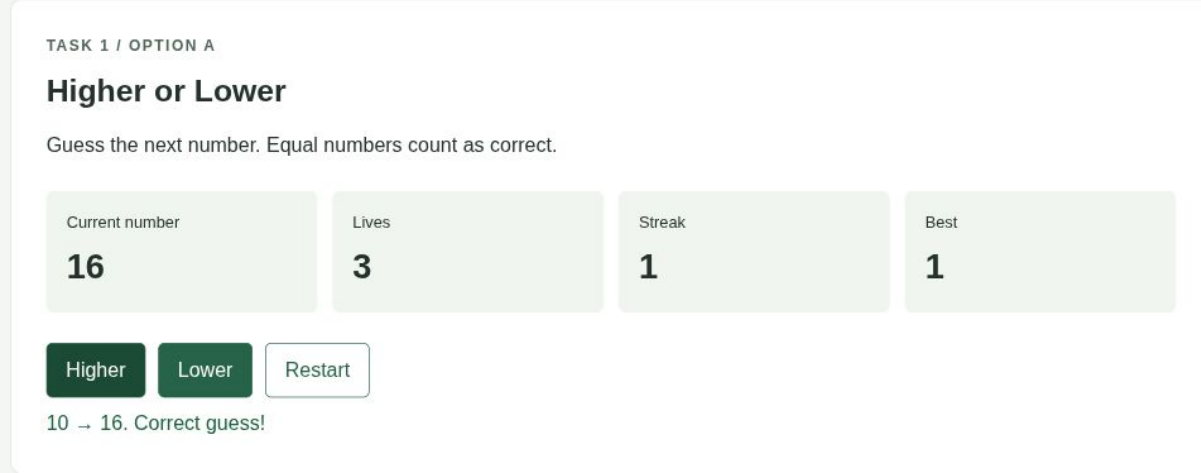


Figure 1. Game state after a correct guess.

Task 2 - Option A: Kanban board

Ten objects store IDs, titles, priorities, tags, completion flags and columns. One render function builds the board. A loop calculates progress. Tag filtering and priority/title sorting change the view. One delegated listener moves cards; the three-card limit counts all tasks, including hidden ones.

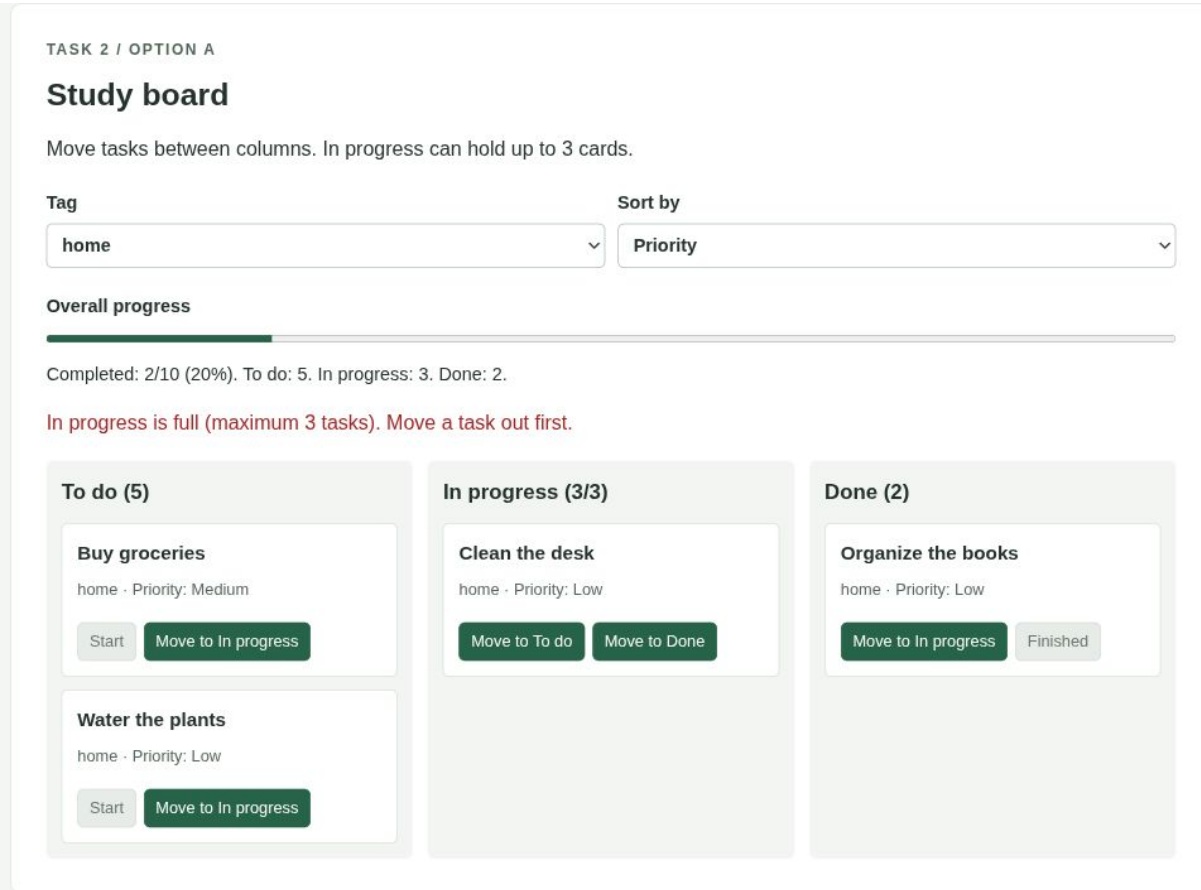


Figure 2. Home filter and a blocked move into the full In progress column.

Task 3 - Option B: Character creator

Seven fields update a live card and points counter. Errors appear after blur and then update on input. Submission counts invalid fields and focuses the first one. Valid data creates a final card using DOM methods, then resets the form. Browser messages are disabled; HTML validation attributes remain.

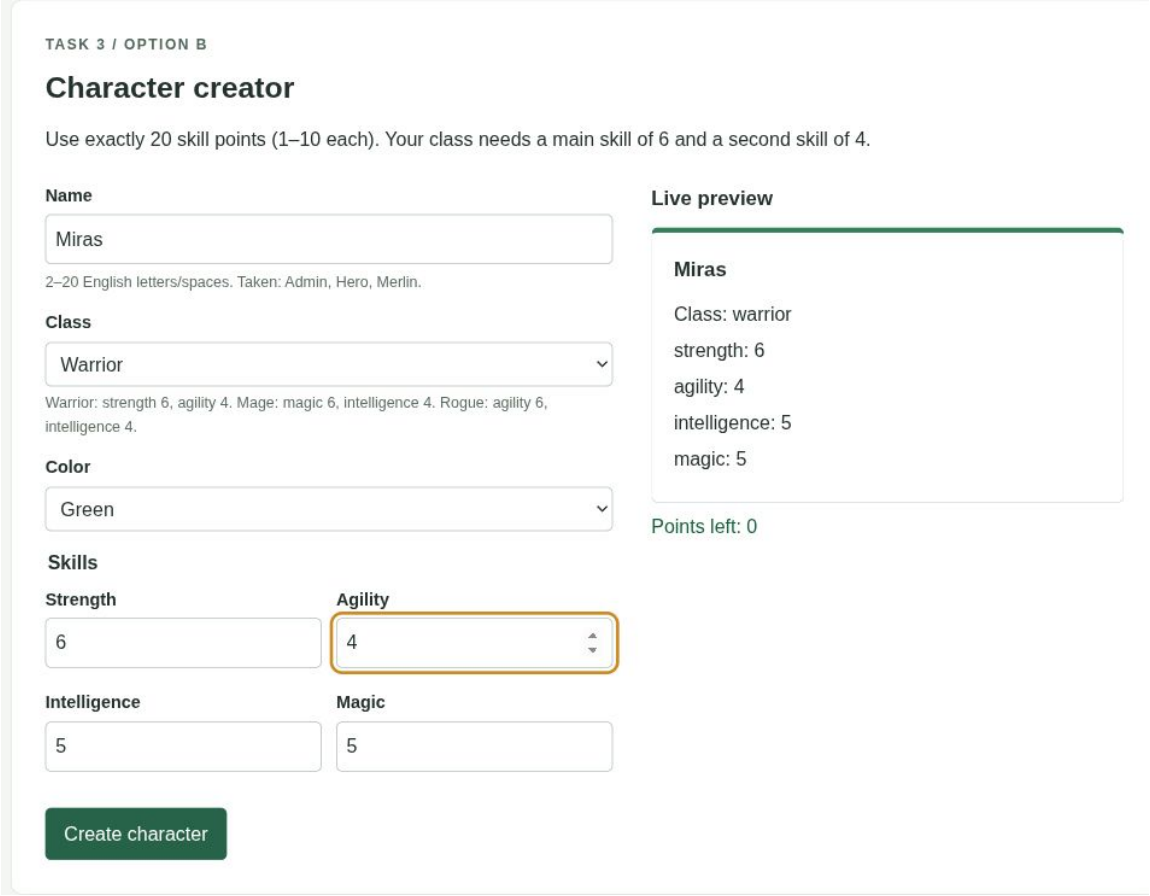


Figure 3. A valid Warrior: 6 + 4 + 5 + 5 = 20 points.

Validation rules

Field	Rule	Valid example	Invalid example
Name	2-20 English letters/spaces; starts with a letter; trimmed name has at least 2 characters.	Miras	M1ras; A
Class / color	Choose one of the available options.	Warrior / Green	Empty selection
Every skill	A whole number from 1 to 10.	5	0; 11; 2.5
Name (JS)	Reject Admin, Hero and Merlin, ignoring case.	Miras	ADMIN
All skills (JS)	The sum must be exactly 20. Order below: strength, agility, intelligence, magic.	6, 4, 5, 5	6, 5, 5, 5
Main skill (JS)	Warrior: strength >= 6. Mage: magic >= 6. Rogue: agility >= 6.	Warrior: 6, 4, 5, 5	Warrior: 5, 4, 6, 5
Second skill (JS)	Warrior: agility >= 4. Mage/Rogue: intelligence >= 4.	Warrior: 6, 4, 5, 5	Warrior: 6, 3, 6, 5

Conclusion

The main lesson is to keep data in variables and arrays, update the DOM from that data, and handle user events with functions. Validation combines HTML constraints with JavaScript rules. The page was checked at 375 px and 1280 px, with no horizontal overflow or console errors.